

RELEVANT JOB HISTORY

- **University of Lausanne** (Lausanne, Switzerland)
Laboratory Assistance for LA-ICP-MS/SIMS Preparation, *2019-2022*
Teaching Assistant for Metamorphic Petrology, *Fall 2018, 2019, 2020*
Teaching Assistant for Geology for EPFL, *Fall 2019, 2020*
Field Assistance BSc Switzerland, *June 2018, 2021*
- **Vanderbilt University** (Nashville, Tennessee, USA)
Research Assistant I – Term Technical, *August 2017 – February 2018*
Teaching Assistant for Petrology, *Spring 2017*
EES Department Analytical Lab Manager, *Fall 2016*
Research Assistant to Calvin Miller, *Spring 2016*
Teaching Assistant for Dynamic Earth, *Fall 2015*
- **Western Kentucky University** (Bowling Green, Kentucky, USA)
Teaching Assistant for Analytical Techniques, *Fall 2013*
Teaching Assistant for Petrology, *Spring 2014*

EDUCATION

- **Ph.D., University of Lausanne**, Lausanne, Switzerland
Institute of Earth Sciences, PhD, *December 2022*
Thesis: On the Crustal Origin of Felsic Volcanic Rocks from the Chon Aike Silicic Large Igneous Province (Patagonia)
Research Advisor(s): Benita Putlitz and Lukas Baumgartner
- **M.S., Vanderbilt University**, Nashville, Tennessee, United States
Earth and Environmental Science, Master of Science, *August 2017*
GPA, 3.84 (4.0 scale)
Thesis: Textural and Geochemical Investigation of the Peach Spring Tuff CA-NV-AZ, USA: Evidence for Dynamic Pre-Supervolcanic Processes
Research Advisor: Calvin Miller
- **B.S., Western Kentucky University**, Bowling Green, Kentucky, United States
Geology Professional Extended Major, Bachelor of Science, *May 2015*
3.69 Overall GPA, 3.70 Major GPA (4.0 scale)
Research Advisor: Andrew Wulff

HONORS AND AWARDS

- Vanderbilt Graduate School Travel Grant (2016-2017)
- Judson Roy Griffin Outstanding Senior Award in Geology (2014-2015)
- Western Kentucky University Alumni Scholarship (2010-2011)

OUTREACH

- Coordinator and organizer for the *Conference Universitaire de Suisse Occidentale – Doctoral Program in Mineral Science* PhD course titled “Introduction to Atom Probe Tomography with Highlights in Geoscience Applications”
Two-day workshop, *Spring 2022*
Invited Speakers: Steven Reddy and David Saxey
- Coordinator and organizer for the *Conference Universitaire de Suisse Occidentale – Doctoral Program in Mineral Science* PhD course titled “Magma Chamber Simulator: Modeling Crustal Geochemical Processes”
Three-day workshop, *Fall 2021*
Invited Speakers: Wendy Bohrsen and Frank Spera

PUBLICATIONS

Under review/In preparation

- [REDACTED] Gerstl, S.S.A., Bloch, E.M., Putlitz, B., Baumgartner, L.P.
“Integrated Textural and Geochemical Analysis of Igneous Zircon and their Xenocrystic Cores by Atom Probe Tomography”
* *in preparation*
- [REDACTED] Gaynor, Sean P., Putlitz, Benita, Baumgartner, Lukas B.
“High precision U-Pb geochronology of an ignimbrite flareup in a silicic large igneous province (Chon Aike, Patagonia)”
* *in preparation for Earth and Planetary Science Letters*
- [REDACTED] Putlitz, Benita, Ulyanov, Alexey, Siron, Guillaume, Baumgartner, Lukas P.
“Large Volumes of Felsic Volcanism by Crustal Melting: tracking crustal contributions in the Patagonian Chon Aike Silicic Large Igneous Province”
**in preparation for Contributions to Mineralogy and Petrology*
- [REDACTED], Putlitz, Benita, Bégué, Florence, Siron, Guillaume, Baumgartner, Lukas P.
“Generating large volumes of crustal-derived high $\delta^{18}\text{O}$ rhyolites in the Chon Aike Silicic Large Igneous Province, Patagonia”
**accepted with minor revisions: Geosphere Themed Issue: “Different Personalities of Granites and Rhyolites: Silicic Magmas from the Lower Crust to the Surface”*

**by labeling a study as in preparation, I am stating that all field work, data collection and modeling are complete, and that the only remaining task prior to submission is completion of the manuscript. Ongoing projects not at this stage are not listed here.*

Printed/In Press

- [REDACTED] Guillermin, Zoé, Putlitz, Benita, Baumgartner, Lukas P. (2022)
“Zircon U-Pb Ages in El Chaltén Jurassic Rhyolites (Chon Aike Silicic LIP, Argentina) Reveal Xenocrystic Remnants of an Unexposed Lower Basement of the Paleo-Pacific Gondwana Margin”
Journal of South American Earth Sciences, 114:103690
- [REDACTED] Miller, Calvin F., Gualda, Guilherme A.R. (2020)
“Architecture of a Super-Sized Magma Chamber and Remobilization of its Basal Cumulate (Peach Spring Tuff, USA)”
Journal of Petrology, 61(1) doi:10.1093/petrology/egaa020

INVITED TALKS

- University of Geneva, 9 March 2022, "Detailed High-Precision Zircon U-Pb Geochronology of a Silicic LIP (Chon Aike, Southern Patagonia): Evidence for Ignimbrite Flare-Ups in Crustal Derived Magmas"
- University of Southern California, Lithospheric Dynamics Seminar, 23 February 2021, "Large Volume Production of High $\delta^{18}\text{O}$ Rhyolites across Space and Time (Chon Aike Silicic Province, Patagonia: An Oxygen Isotope Investigation using in-situ SIMS and LF methods"

PROFESSIONAL EXPERIENCE

Workshops

- Granites II Summer School
Roscoff, France, *July 2019*
Organizing Committee: Laurent Arbaret, Fabrice Gaillard, Erwan Hallot
- International Association of Volcanology and Chemistry of the Earth's Interior, Workshop Sicily and Aeolian Islands, Italy, *July 2016*
- Secondary Ion Mass Spectroscopy Workshop
University of California at Los Angeles, Los Angeles, United States. *February 27 – March 3, 2017.*
Leaders: Kevin McKeegan, Mark Harrison, Elizabeth Bell, Melanie Barboni, Haolan Tang

Conferences

- Geological Society of America, Annual Conference, Denver, Colorado, United States, *October*
[REDACTED]
- [REDACTED]
[REDACTED] in the El Quemado Complex of the Chon Aike Silicic Large Igneous Province: A Short High-Flux Event within a Longer-Lived SLIP.
- Geological Society of America, Annual Conference, Portland, Oregon, United States, *October 2021. Oral*
[REDACTED], Putlitz, B., Baumgartner, L.P., Ulyanov, A., Siron, G.
"Isotope Characteristics of a Large Silicic Province: Crustal versus Mantle Processes in the Generation of the Jurassic Chon Aike Silicic Province (Argentina)"
Geological Society of America *Abstracts with Programs*. Vol 53, No. 6
- [REDACTED], Benita, Baumgartner, Lukas, Guillermin, Zoé, Bégué, Florence
"Magmas of the El Quemado Complex (Chon Aike Silicic Igneous Province, Patagonia: Elevated Oxygen Isotope Signatures across Space and Time"
22nd EGU General Assembly, id.21265
- [REDACTED]
[REDACTED] Miller, Calvin F., Gualda, Guilherme A.R., Bradshaw, Richard W.

"Complex Reheating and Reactivation of Basal Cumulate in the Peach Spring Tuff Magma Body Revealed by Glass and Feldspar Textures and Compositions"
AGUFM, V33D-0271

- Geological Society of America, Annual Conference, Seattle, Washington, United States. *October 2017. Poster*
[REDACTED]
"Heterogeneous Glasses Preserve Chemical and Dynamic Processes in Partially Remelted and Remobilized Cumulate of the Peach Spring Tuff (CA-NV-AZ), US"
Geological Society of America *Abstracts with Programs*. Vol 49, No. 6
- Geological Society of America, Cordilleran Section Conference, Honolulu, Hawaii, United States. *May 2017. Oral*
[REDACTED], Miller, Calvin M., Gualda, Guilherme AR.
"Heterogeneities in pumice and fiamme provide insights into thermal and mechanical history of partially remelted and remobilized crystal mush in the magma chamber that fed the Peach Spring supereruption (AZ, USA)"
Geological Society of America *Abstracts with Programs*. Vol 49, No. 4
- Geological Society of America, Annual Conference, Denver, Colorado, United States. *October*
[REDACTED], Miller, Calvin M., Ferguson, Charles A., Barry, Erin
"Peach Spring Tuff (AZ, USA): Implications of Zonation for the Supereruption and Reconstruction of Chamber Stratigraphy"
Geological Society of America *Abstracts with Programs*. Vol 48, No. 7
- Geological Society of America, Annual Conference, Vancouver, British Columbia, Canada.
[REDACTED], L., Miller, Calvin M., Wulff, Andrew H., McDowell, Susanne M., Covey, Aaron K.
"Dynamics of Destabilization: Pumiceous Evidence of Reheated, Remelted, and Remobilized Cumulate from the Base of a Supervolcano Magma Chamber"
Geological Society of America *Abstracts with Programs*. Vol 46, No. 6. p. 510

FIELD AND ACADEMIC RESEARCH EXPERIENCE

- Vanderbilt University Research Experience for Undergraduates: Before and After a Supervolcano
May – June 2016, Graduate Assistant Coordinator and Instructor
January 2015, Returning Participant (served as instructor, coordinator, "role model")
May – June 2014, Undergraduate Participant
- Centre College (Kentucky) Study Abroad: Physical Science of Volcanoes, New Zealand
January 1 – January 21, 2016
Leaders: Joe Workman, Kerry Paumi
Title: Graduate Teaching Assistant

ANALYTICAL METHODS EXPERIENCE

Sample Preparation

- Preparation of Rock Thin Sections: Ultramicrotome Thin Sections
- Crushing and Panning for Size Fractions
- Frantz Magnetic Separation
- Heavy Liquid Mineral Separation
- Epoxy Rock Chip Mounts
- Powder and Fusing Glass Disks for X-Ray Fluorescence

Analytical Facilities

- Petrographic Light Microscope
- Raman Spectroscopy
- Single Crystal and Powder X-Ray Diffraction
- Transmitting Electron Microscope (TEM)
- Scanning Electron Microscope (SEM)
 - Secondary Electrons (SE)
 - Energy Dispersive Spectroscopy (EDS)
 - Cathodoluminescence (CL)
 - Electron Backscatter Diffraction (EBSD)
- Electron Microprobe Analyzer (EMPA)
- Laser Ablation – Inductively Coupled Plasma – Mass Spectroscopy (LA-ICP-MS)
- X-ray Computed Tomography (XCT)
- Secondary Ion Mass Spectroscopy (SIMS)
- Mineral Liberation Analysis (MLA)
- Focused Ion Beam Scanning Electron Microscope (FIB-SEM)
- Atom Probe Tomography (APT)

ADDITIONAL QUALIFICATIONS

- Experienced with both Microsoft and Macintosh Programs
- Microsoft Office (Word, Excel, PowerPoint)
- Adobe Suite
- ArcGIS, QGIS
- TOPAS, MagicPlot
- Rhyolite-MELTS, MELTS-Barometry
- Programming Languages: IDL, LaTeX, Python
- Data Reduction Programs: Glitter, Iolite, Lamtrace